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Vision Global Proposal Automation System

Buyer-to-Supplier Procurement Automation Platform

PREPARED FOR	Techspire
DOCUMENT	Project Proposal & Solution Overview
DATE	26 October 2026
VERSION	1.0

1. Executive Summary

Vision Global is a full-stack procurement automation platform that connects buyers and suppliers through a single, intelligent workflow. It removes the manual effort traditionally involved in collecting supplier quotations, pricing them, routing them to the right buyers, and following up — turning a slow, email-and-spreadsheet process into a streamlined, auto-driven pipeline.

The system automatically ingests supplier quotations from Email and WhatsApp, parses each machine and its price into structured rows, attaches supplier-sent photos, applies a configurable profit margin, and dispatches tailored quotations to segmented buyers over their preferred channel. Scheduled follow-ups and a unified chat history ensure no opportunity is lost. The result is faster turnaround, fewer errors, and complete visibility across the buyer–supplier lifecycle.

Highlight	What it delivers
Zero manual entry	Supplier quotes are fetched & parsed automatically by keyword
Multi-channel	Email (Gmail) and WhatsApp (Meta Cloud API), send and receive
Smart routing	Quotations go only to buyers who enquired, filtered by segment
Always following up	Daily / weekly automated follow-ups with full chat history

2. Business Problem & Objectives

The challenge

Procurement teams spend hours manually reading supplier emails, copying machine names and prices into spreadsheets, calculating margins, identifying which buyers to contact, and chasing responses. This manual process is slow, error-prone, and difficult to track — quotations get missed, follow-ups are forgotten, and there is no single record of what was sent to whom.

Objectives

- **Eliminate manual data entry** — capture supplier quotes directly from the inbox and WhatsApp.
- **Standardise pricing** — apply a consistent, editable margin to every machine.
- **Target the right buyers** — match quotations to buyers by machine and interest segment.
- **Automate follow-ups** — keep deals moving without human reminders.
- **Centralise communication** — one chat history per buyer across all channels.
- **Give full visibility** — a live dashboard of enquiries, proposals and quotations.

3. Solution Overview

Vision Global is delivered as a modern web application with a clean, orange-themed interface and a secure Python backend connected to a live MySQL database. The platform is organised into five core modules that mirror the real procurement workflow:

Module	Purpose
Dashboard	At-a-glance KPIs — enquiries, proposals, quotations and activity.
Buyer Enquiry	Captures buyer requirements (machine, quantity, channel), auto & manual.
Supplier Proposal	One row per machine from each supplier quote — price, photos, margin, buyer segment.
Buyer Quotation	Quotations sent to buyers, with channel, message preview, follow-up and chat history.
Administration	Users, keywords, channels, message templates and integrations.

Design principles

- **Automation-first** — the system does the work; staff review and approve.
- **Single source of truth** — every record lives in the live database, never mock data.
- **Channel-aware** — the right template and styling for Email vs WhatsApp.

4. Key Modules in Detail

Supplier Proposal — the automation core

Supplier quotations arriving by Email or WhatsApp are matched against configured keywords and parsed into one editable row per machine. Each row shows the supplier, machine name, parsed price, any attached photos, an editable profit margin, the computed final price, and a buyer segment selector.

- **Multi-machine quotes** become separate rows — image 1 to machine 1, image 2 to machine 2.
- **“Get Photo”** emails the supplier; their reply photos auto-attach to the right row.
- **Buyer segment** (Everyone / Cold / Hot / Very Hot) controls who receives the quotation.
- **Send** dispatches to only the buyers who enquired for that machine and match the segment.

Buyer Quotation — close the loop

- **Channel column** shows how each quotation was delivered (Email / WhatsApp).
- **View** renders the exact message sent, from the channel-specific template.
- **Follow-up** — Daily, Weekly or Don’t-Disturb, sent automatically on schedule.
- **Chat history** — quotation, follow-ups and buyer replies in one styled thread.

Administration — full control

- **Keywords** drive what gets pulled from the inbox and WhatsApp.
- **Templates** — four message templates: Buyer Quotation and Follow-up, each in Email & WhatsApp.
- **Integrations** — live status for Email and WhatsApp, with webhook configuration.
- **Users & Channels** — manage access and the communication channels in use.

5. Automation Capabilities

Capability	How it works
Email intake	A background scheduler polls the inbox every 30 seconds and imports keyword-matched supplier quotes.
WhatsApp intake	Meta Cloud API webhook delivers supplier messages instantly; keyword-matched quotes become proposal rows.
Machine & price parsing	Each quote is parsed into machine name + price, handling multiple machines and varied formats.
Photo handling	Supplier image attachments are saved and shown in the Photos column; reply photos auto-attach.
Margin & pricing	An editable margin per row computes the final buyer price automatically.
Segmented send	Quotations reach only matching buyers (by machine + segment), over their channel.
Automated follow-ups	Daily (24h) or weekly cadence, sent and logged without manual action.
Buyer replies	Replies on Email and WhatsApp are captured into the buyer’s chat history.

6. Technical Architecture

Technology stack

Frontend	React 18 + Vite — responsive single-page application
Backend	Python FastAPI (REST API) with Uvicorn
Database	MySQL — two schemas: Etex_admin (config) & Etex_Proposal (operations)
Authentication	JWT-based sessions with bcrypt-hashed passwords
Email	Gmail over SMTP (send) and IMAP (fetch), via App Password
WhatsApp	Meta WhatsApp Cloud API — send, follow-ups, inbound webhook
Scheduling	Background worker for auto-fetch, follow-ups and reply ingestion

How the pieces connect

- The React frontend talks to the FastAPI backend over a secure REST API.
- The backend reads and writes the live MySQL databases — all data shown is real, never mock.
- Email and WhatsApp integrations feed supplier quotes in and send buyer quotations out.
- A scheduler runs continuously to fetch, parse, follow up and record replies.

7. End-to-End Workflow

1. **Buyer enquiry.** A buyer's requirement (machine, quantity, channel) is captured in the system.
2. **Supplier quote arrives.** A supplier sends a quotation by Email or WhatsApp containing machine name(s) and price(s).
3. **Auto-capture & parse.** The system matches keywords and creates one proposal row per machine, with price and photos.
4. **Price & segment.** Staff review the margin and choose the buyer segment for each machine.
5. **Send quotation.** The quotation is sent to matching buyers over their channel, using the correct template.
6. **Follow up.** Daily or weekly follow-ups are sent automatically until the buyer responds.
7. **Track replies.** Buyer replies land in the chat history, giving a complete conversation record.

8. Business Benefits

Benefit	Impact
Faster turnaround	Quotes are captured and routed in seconds, not hours.
Fewer errors	Automated parsing removes copy-paste mistakes in machine names and prices.
Higher conversion	Automated follow-ups keep every opportunity alive.
Better targeting	Segment-based sending reaches the buyers most likely to purchase.
Full visibility	Live dashboard and per-buyer chat history across all channels.
Lower effort	Staff shift from data entry to review and decision-making.

9. Security & Data

- JWT-secured API access with bcrypt-hashed credentials.
- Secrets (database, email and WhatsApp keys) kept in server-side configuration, never in the browser.
- All operational data persisted in the organisation’s own MySQL databases.

10. Roadmap & Next Steps

Delivered

- Five-module web application on live MySQL data.
- Email intake, parsing, photo handling and segmented sending.
- WhatsApp Cloud API integration — send, follow-ups and inbound webhook.
- Automated follow-up scheduler and unified chat history.

Next

- Approved WhatsApp message templates for business-initiated (cold) outreach.
- Production hosting with a permanent public endpoint for the WhatsApp webhook.
- Analytics & reporting on conversion and supplier performance.
- Role-based access and audit trails for larger teams.

Conclusion. Vision Global replaces a manual, fragmented procurement process with a single automated platform that captures supplier quotes, prices them, routes them to the right buyers and follows up — end to end. It is already live against real data and integrated with Email and WhatsApp, with a clear path to full production deployment.

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